1. Program Overview

Capstone Introduction

Capstone 2 is an advanced, industry-style data analytics simulation. Student teams solve real-world sectoral problems using Python for the full ETL pipeline and Tableau for visualization and storytelling. The objective is to demonstrate end-to-end analytical capability with a focus on code quality, reproducibility,and decision-oriented insight delivery.
Each team selects a sector and a problem statement, then executes a complete analytics workflow —dataset sourcing, Python-based ETL, exploratory and statistical analysis, Tableau dashboard design, and final business recommendations. All work is version-controlled via a public GitHub repository using a structured branching and pull request workflow.

2. Team Structure & Roles

Team Size: 5 Members
Formation: Teams are formulated by students to ensure balanced work allocation. No swapping is permitted.

3. Execution Schedule (Day-by-Day)

Week 1: Data Sourcing, ETL & Cleaning

Objective: Secure mentor approval and prepare a clean, analysis-ready dataset committed to GitHub.

Day 1–2: Discovery & Setup

• Task: Meet your team, assign roles, select a sector (Retail, Finance, Health, etc.), and find a dataset.
• Task: Create the GitHub repository using the naming convention SectionName_TeamID_ProjectName and set up the folder structure.
• Rule: Dataset must be raw (min 5,000 rows, 8 columns). No pre-cleaned Kaggle competition files.

Day 3: Gate 1 — The "Go/No-Go" Checkpoint

• Action: Submit your Gate 1 proposal — problem statement, dataset link, and initial data dictionary.
• Critical: You cannot proceed to deep analysis until the mentor approves your dataset.
• Backup: Have at least two backup datasets ready from the same sector in case the primary is rejected.

Day 4–5: ETL Pipeline & Cleaning

• Task: Commit the raw dataset to data/raw/ — never edit this file.
• Task: Build the Python cleaning pipeline in notebooks/02_cleaning.ipynb. Log every transformation
step.
• Task: Output cleaned data to data/processed/ and commit all notebooks to GitHub.
• Deliverable: A fully standardised dataset committed to GitHub, ready for analysis.

Week 2: Analysis, Dashboarding & Reporting
Objective: Uncover insights, build the Tableau dashboard, and deliver the final submission.

Day 6–7: EDA & Statistical Analysis
• Task: Run exploratory data analysis in notebooks/03_eda.ipynb — trends, distributions, outliers.
• Task: Perform statistical analysis in notebooks/04_statistical_analysis.ipynb — correlation, regression, hypothesis testing as applicable.
• Task: Compute KPIs and create final data load in notebooks/05_final_load_prep.ipynb.

Day 8–9: Tableau Dashboard
• Task: Build the final dashboard on Tableau Public. Design for decision-making, not decoration.
• Requirement: Must include interactive filters. Do not hard-code numbers.
• Task: Publish the dashboard, copy the public URL, and commit screenshots to tableau/screenshots/.

Day 10: Final Submission
• Task: Draft 3–5 specific, data-backed business recommendations.
• Task: Complete the final project report (PDF) and presentation deck.
• Submission: Ensure all assets are in the GitHub repo and all links work before the deadline.

4. Technical Requirements & Tooling

Tool	Status	Notes
Python + Jupyter	Mandatory	All ETL, cleaning, and statistical analysis must be in Python using Jupyter Notebooks (.ipynb files).
Google Colab	Supported	Teams may use Colab as their notebook environment. All .ipynb files must be committed to GitHub.
GitHub	Mandatory	All work must be version-controlled. Every member must have commits and PR contributions.
Tableau Public	Mandatory	All dashboards must be built and published on Tableau Public. Share the public URL and screenshots.
SQL	Optional	May be used for initial data extraction only. All SQL logic must be documented in the repo.

GitHub Repository Structure
All teams must maintain the following folder structure. Commit all notebooks, scripts, and reports here.

```
SectionName_TeamID_ProjectName/
- README.md
- data/
    - raw/        ← Original dataset (never edited)
    - processed/  ← Cleaned output from pipeline
- notebooks/
    - 01_extraction.ipynb
    - 02_cleaning.ipynb
    - 03_eda.ipynb
    - 04_statistical_analysis.ipynb
    - 05_final_load_prep.ipynb
- scripts/
    - etl_pipeline.py
- tableau/
    - screenshots/
    - dashboard_links.md
- reports/
    - project_report.pdf
    - presentation.pdf
- docs/
    - data_dictionary.md
```

GitHub Contribution Audit
Warning: The faculty will review GitHub Insights and PR history for every member. If a team member has
no visible commits or pull requests, they may face an individual mark deduction.

5. Dataset Requirements

Minimum Standards
• Row count: At least 5,000 rows
• Column count: At least 8 meaningful analytical columns (numerical or categorical — not IDs or free text)
• Quality: Must contain realistic issues — missing values, inconsistent formats, or category variations that require cleaning
• Format: Tabular, row-level records suitable for Python ETL, KPI design, and Tableau visualisation

Approved Dataset Sources
• https://datasetsearch.research.google.com/
• https://www.kaggle.com/datasets (raw datasets only — see Kaggle rules below)
• https://data.gov.in/
• https://dataverse.harvard.edu/
• https://microdata.worldbank.org/
• https://archive.ics.uci.edu/
• https://data.world/ | https://datahub.io/

Special Rules for Kaggle Datasets

Kaggle contains both raw datasets and highly polished competition datasets. Most trending or competition-ready files are already clean and are not suitable for this capstone.
When using Kaggle, ensure:
• The dataset is raw or minimally processed
• Row-level records are available
• Missing values and inconsistencies are present
• No pre-built dashboard or feature-engineered version is used
Search using keywords such as: raw, original, transactional, logs, survey, records, events.
Note: Mentors may reject Kaggle datasets that are already analysis-ready.

Backup Dataset Requirement
Each team must submit one primary dataset and keep at least two backup options ready from the same
sector. If the primary dataset is rejected at Gate 1, switch to a backup without delaying the timeline.

6. Submission Checklist

Before the final deadline, ensure your submission includes all of the following:

Asset 1: GitHub Repository
• Public repository created with correct naming convention
• All notebooks committed in .ipynb format
• data/raw/ contains the original, unedited dataset
• data/processed/ contains the cleaned output
• tableau/screenshots/ contains dashboard screenshots
• tableau/dashboard_links.md contains the Tableau Public URL
• docs/data_dictionary.md is complete
• README.md explains the project, dataset, and team
• All members have visible commits and pull requests

Asset 2: Tableau Dashboard
• Dashboard published on Tableau Public (accessible via public URL)
• At least one interactive filter included
• Dashboard addresses the business problem directly
• URL shared in dashboard_links.md and in the final report

Asset 3: Project Report (PDF) — 10 to 15 Pages
• Cover page with title, sector, team details, and institute
• Executive summary (1 page)
• Sector context and problem statement
• Data description — source, structure, size, and limitations
• Data cleaning and ETL methodology (Python pipeline)
• KPI and metric framework
• EDA with visualisations and written insights
• Statistical analysis results
• Dashboard design — screenshots and explanation
• 8–12 key insights written in decision language

• 3–5 actionable business recommendations
• Impact estimation, limitations, and future scope
• Contribution matrix (must match GitHub history)

7. Evaluation Rubric (100 Marks)

Area	Marks	Focus
Problem Framing	10	Is the business question clear and well-scoped?
Data Quality & ETL	15	Is the Python pipeline thorough and documented?
Analysis Depth	25	Are statistical methods applied correctly with insight?
Dashboard & Visualization	20	Is the Tableau dashboard interactive and decision-relevant?
Business Recommendations	20	Are insights actionable and well-reasoned?
Storytelling & Clarity	10	Is the doc professional & coherent?

8. What Makes a High-Scoring Project

• A clear, well-scoped business problem statement
• A documented, reproducible Python ETL pipeline in Jupyter Notebooks
• Strong statistical analysis — not just counts and averages
• A Tableau dashboard that answers real business questions
• Insights written in decision language, not chart descriptions
• Practical, specific recommendations with estimated business impact
• A professional, coherent presentation deck and final report
• Every team member with visible GitHub contributions

9. Rules & Academic Integrity

• No plagiarism — all analysis, code, and recommendations must be your team's original work
• No copying from other teams — projects are compared during evaluation
• Every team member must understand the dataset, cleaning steps, KPIs, and dashboard structure
• Viva questions can be directed to any team member — not just the presenter
• Late submission will incur a grade penalty
• Copied dashboards, reused templates without understanding, or duplicate work across teams will be penalised

Contribution Matrix: A contribution matrix must be included in the final report and must match the evidence in GitHub Insights, PR history, and committed files.

10. How This Helps Your Career

• This becomes your primary interview project — a real end-to-end analytics case
• Demonstrates Python-based ETL, statistical analysis, and Tableau skills in one portfolio piece
• Your GitHub repository is a live portfolio — keep it clean and well-documented
• The Tableau Public dashboard is publicly accessible — share the link on LinkedIn and your resume
• Shows real business thinking — the ability to convert data into decisions
• Improves your confidence in explaining data, methodology, and recommendations under pressure

Tools don't get you marks. Thinking gets you marks.
Dashboards don't get you marks. Decisions get you marks.

Data Visualization & Analytics

Final Project Report

https://docs.google.com/document/d/1-1TUNkeGw0X3C3nQ6wVi18i1oQnpDgjSK6dmignS-Nw/edit?tab=t.0

Project Flow.

PROJECT WORKFLOW

Project Flow.

01

Define the problem

Select the sector, define the problem statement, and identify the business question.

02

Prepare the data

Load, inspect, clean, filter, group, and merge data in Python.

03

Analyze the data

Run descriptive statistics, correlations, outlier checks, and deeper comparisons.

04

Build the dashboard

Use Tableau Public to convert analysis outputs into interactive charts and dashboards.

05

Write recommendations

Translate the analysis into clear, decision-oriented recommendations.

06

Submit and defend

Package the repo, report, PPT, dashboard link, and prepare for viva.

How will Capstone 2 be Evaluated?

Official 100-Mark Rubric

- **Problem Framing — 10 marks** Clear, well-scoped business question.
- **Data Quality & ETL — 15 marks** Documented and reliable Python pipeline.
- **Analysis Depth — 25 marks** Correct statistical methods with real insight.
- **Dashboard & Visualization — 20 marks** Interactive, decision-relevant Tableau Public dashboard.
- **Business Recommendations — 20 marks** Actionable and well-reasoned recommendations.
- **Storytelling & Clarity — 10 marks** Coherent report, deck, and walkthrough.

What reviewers will look for

- **Use evidence, not volume** Marks are awarded for analytical thinking and decision relevance.
- **Keep all assets aligned** The repo, dashboard, report, and PPT should tell the same story.
- **Show visible team contribution** Every member should have real GitHub commits and PR activity.
- **Write practical recommendations** Recommendations should come directly from the analysis.